JOBGENIE – AI POWERED JOB PORTAL

Project Synopsis

Maharishi Markandeshwar (Deemed to be University)

In partial fulfillment of the requirement for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

By:

Ishika Bansal (11222567)

Computer Science & Engineering Department,

M.M. Engineering College, Mullana, Ambala – 133207, Haryana, India

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CHAPTER 1

Introduction

JobGenie is an advanced AI-powered job portal developed to bridge the gap between job seekers and recruiters through an intelligent, fully automated, and highly scalable digital ecosystem. As recruitment becomes increasingly competitive, traditional hiring methods struggle to keep up with the growing volume of applicants and the rising demand for specialized skills. JobGenie addresses these issues by offering a streamlined job search process driven by machine learning, NLP, and data analytics.

This platform provides a dynamic environment where job seekers can create comprehensive profiles, upload resumes, receive personalized job recommendations, and track their applications in real-time. Meanwhile, recruiters can post openings, screen applicants using AI-driven compatibility scoring, and manage the hiring pipeline from a centralized dashboard. With automated notifications, real-time updates, and smart ranking algorithms, JobGenie reduces manual workload and ensures a faster, more efficient recruitment cycle.

Developed using the MERN stack, JobGenie ensures fast load times, responsive user interfaces, robust authentication, and modular architecture. The platform integrates resume parsing, role-based dashboards, and candidate–recruiter communication to create a seamless digital hiring experience. As employment trends shift towards remote, hybrid, and global opportunities, systems like JobGenie make hiring more accessible, transparent, and intelligent.

CHAPTER 2

Objectives, Features

Objectives:

The primary objective of JobGenie is to revolutionize the hiring process by integrating automation and artificial intelligence into every stage of recruitment. The system aims to enhance user experience, reduce manual tasks, and enable accurate job-candidate matching.

• Provide accurate, AI-driven job recommendations based on candidate skills, experience, and preferences.
• Automate resume parsing to extract relevant information including skills, education, and work history.
• Build a recruiter-friendly dashboard for managing job posts, applications, and shortlisting.
• Ensure transparency through real-time notifications and status updates.
• Enable secure authentication and protect user data using encryption.
• Offer intuitive UI/UX to improve accessibility and user engagement.
• Reduce hiring bottlenecks by ranking candidates based on AI-generated match scores.
• Provide data-driven insights through dashboards and analytics.

Features:

• Separate dashboards for job seekers, recruiters, and admins.
• Resume parsing and skill extraction using NLP.
• Personalized job recommendation feed and job alerts.
• Candidate ranking and job-candidate compatibility scoring.
• Smart search with multi-parameter filters.
• Real-time email and platform notifications.
• Automated pipeline for job tracking, shortlisting, and communication.
• Admin controls for system monitoring, user management, and reports.

CHAPTER 3

Feasibility Study, Future Enhancements, Target Audience

Feasibility Study:

To ensure sustainability and long-term viability, JobGenie underwent a detailed feasibility analysis across technical, operational, and economic parameters.

Technical Feasibility:

The MERN stack provides a strong foundation for building fast, scalable applications. React improves front-end responsiveness, while Node.js ensures smooth backend operations. MongoDB supports flexible data structures needed for resumes, job descriptions, and user profiles. AI and NLP modules enhance automation and accuracy in job matching.

Operational Feasibility:

The platform is highly user-friendly and requires minimal technical knowledge. Its clear navigation, role-based dashboards, and automated workflows ensure smooth operation. Recruiters can manage large applicant pools without effort, while job seekers benefit from smart job suggestions and guided application flows.

Economic Feasibility:

JobGenie uses cost-effective cloud infrastructure and free/open-source development tools. Businesses adopting the platform can significantly reduce hiring expenses through automation, decreasing dependency on external recruitment agencies.

Future Enhancements:

- AI chatbot for personalized career guidance.
- Interview scheduling automation.
- Voice-enabled resume builder.
- Mobile app for real-time alerts.
- Deep-learning–based personality & culture fit analysis.
- Integration with LinkedIn, Indeed, and Naukri.
- Video interview analysis using AI for soft-skill evaluation.

Target Audience:

- Students and fresh graduates.
- Working professionals seeking job transitions.
- Companies and HR teams needing automated hiring.
- Placement cells and training departments.
- Recruiters hiring for large-scale positions.

CHAPTER 4

Methodology / Planning of Work

Methodology:

JobGenie follows a structured development approach to ensure reliability, modularity, and user satisfaction. The methodology includes research, requirement analysis, design, development, testing, deployment, and maintenance phases.

Steps Followed:

• Requirement gathering and identification of existing hiring challenges.
• Research on AI-based job matching and resume parsing.
• UI/UX design using wireframes and user flow diagrams.
• Database planning and schema creation.
• Building authentication modules and REST APIs.
• Integrating AI/NLP engine for resume and job analysis.
• Frontend development using ReactJS and Tailwind CSS.
• System testing, debugging, and performance optimization.
• Deployment using cloud platforms.
• Continuous updates, monitoring, and improvements.

Data Collection & Analysis:

The system collects job descriptions, resumes, and user activity logs. This data is analyzed using ML pipelines for tokenization, semantic similarity checks, and skill extraction, ensuring accurate and user-specific recommendations.

Planning of Work:

Development work is divided into modules: authentication, dashboard creation, AI engine, job posting system, application tracking, notifications, and admin features. Each module is built, tested, and integrated during iterative cycles.

CHAPTER 5

Software / Hardware Requirements

Software Requirements:

Frontend: ReactJS, TailwindCSS
Backend: Node.js, Express.js
Database: MongoDB Atlas
Cloud Storage: Cloudinary, Firebase Storage
AI Tools: Python NLP libraries, TensorFlow/PyTorch (optional)
Other Tools: VS Code, Git, Postman

Hardware Requirements:

Any standard computing device (laptop/desktop/tablet/mobile) with a stable internet connection. The cloud-based architecture ensures minimal resource requirements and provides seamless access across all devices.

CHAPTER 6

Conclusion

Expected Outcome:

JobGenie is expected to simplify and modernize the job searching and hiring experience through AI automation and intelligent workflows. The platform will deliver faster hiring cycles, accurate job matching, transparent communication, and an organized recruitment process.

Conclusion:

JobGenie transforms traditional recruitment by incorporating AI, NLP, and automation to improve accuracy, reduce effort, and enhance user experience. Its future scalability, modular design, and smart features make it a powerful tool for job seekers, recruiters, and organizations. With continuous enhancements, JobGenie is poised to become a complete AI-driven hiring solution.

References:

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- https://www.kaggle.com
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